
EGA: Adapting Frozen Encoders for Vector Search without Reshaping Unseen-Class Geometry

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Abstract

1 Vector similarity search systems built on frozen vision-language encoders routinely
2 face queries from classes absent at adapter training time: both the indexed database
3 and the live query stream contain categories that were never labeled. We study
4 how lightweight adapters behave under this deployment-induced distribution shift,
5 and find a consistent in-distribution/out-of-distribution (ID/OOD) tradeoff among
6 existing training paradigms. Global contrastive objectives such as ICon and SRL,
7 when paired with high-capacity adapters, achieve near-perfect in-distribution Label
8 Precision on CIFAR-100 (0.999 and 0.992 at $K=1$, $n_{\text{probe}}=1$) but collapse to
9 0.562 and 0.552 on the unseen classes of Food-101, a 44 percentage point drop.
10 Low-capacity alternatives such as LoRA avoid this collapse but underperform
11 substantially in-distribution. We propose Euclidean Geodesic Alignment (EGA),
12 a residual adapter trained with a small-margin triplet loss on the class-aware
13 neighborhood graph of pre-trained embeddings. EGA’s design induces gradient
14 sparsity at convergence (96.5% of training triplets produce zero gradient), which
15 empirically preserves the pre-trained semantic structure for unseen categories while
16 permitting high-capacity in-distribution refinement. Across CIFAR-100 (ID) and
17 three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), the six methods
18 we evaluate occupy distinct operating points in the (ID, OOD) Label Precision
19 plane. EGA reaches the highest in-distribution Label Precision among adapters
20 that avoid OOD collapse (0.705 on CIFAR-100, vs. 0.6176 for capacity-matched
21 LoRA at $r=128$), with bounded OOD degradation (-7 to -9 pp on CIFAR-10 and
22 Food-101, vs. -32 to -33 pp for ICon and SRL). The benefits transfer to stronger
23 backbones: on DINOv2-large and SigLIP, EGA improves Label Precision@1 by
24 4.75 and 8.55 percentage points respectively.

25 1 Introduction

26 A vector similarity search system deployed in production rarely operates on a fixed label space. A
27 retailer indexes product images today and adds a new category line next month; a content platform
28 builds an index over existing tags and continually ingests media from emerging topics; a scientific
29 image database is annotated for known phenotypes and queried for ones discovered later. In such
30 scenarios, the index and the query stream together cover a label space strictly larger than what was
31 available at the time any model was trained or adapted.

32 This deployment reality, in many cases, sits awkwardly with how adapters over frozen vision-
33 language encoders are currently trained. The dominant paradigm takes a labeled subset, optimizes
34 the embedding geometry to separate those classes, and ships the resulting adapter as a drop-in
35 replacement for the frozen encoder’s output. The implicit assumption is that geometry tightened
36 for seen classes will, at worst, leave unseen-class geometry unchanged. We show this assumption

fails systematically: high-capacity adapters trained with global contrastive objectives reshape the entire embedding distribution, pulling unseen-class samples toward whichever seen-class clusters they happen to resemble in the pre-trained metric. The retrieval system inherits this distortion silently: approximate nearest neighbor indices are still well-behaved geometrically, but the neighbors they return belong to the wrong semantic class.

Two design choices in current adapter training are responsible for this distortion. First, the loss function: global contrastive objectives such as ICon [1] and SRL [4] compute gradients over the full pairwise similarity matrix in every step, so seen-class supervision exerts pressure across the entire embedding distribution rather than only on the regions occupied by labeled samples. Second, the adapter capacity: a sufficiently expressive adapter has the representational room to comply with that pressure, and zero-initialized residual designs do not by themselves prevent the cumulative perturbation from reaching unseen-class regions. Existing alternatives address one factor at a time. Low-rank adapters such as LoRA [10] cap the second factor, which avoids the worst out-of-distribution (OOD) failures but limits in-distribution (ID) refinement, while keeping the global loss leaves the first factor untouched. We argue that adapters intended to operate under unseen-class queries should be designed against both factors at once.

The cost of failing on either factor is large. On in-distribution data (CIFAR-100), high-capacity global contrastive adapters achieve near-perfect Label Precision: ICon reaches 0.999 and SRL reaches 0.992 at $K = 1$, $n_{\text{probe}} = 1$. On the strictly unseen classes of Food-101, however, the same adapters collapse to 0.562 and 0.552, which is 44 percentage points below their in-distribution scores and between 32 and 33 percentage points below the frozen CLIP baseline. LoRA variants avoid this collapse but pay for it in-distribution: their Label Precision is bounded between 0.61 and 0.67, well below what a high-capacity adapter can achieve when the loss does not over-extend into unseen-class regions.

We propose Euclidean Geodesic Alignment (EGA), a residual adapter designed to address both factors simultaneously. EGA pairs a high-capacity zero-initialized residual architecture with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings, together with ℓ_2 projection back onto the unit hypersphere. The triplet loss with a small margin induces a self-limiting training dynamic: as the adapter converges on seen classes, the fraction of margin-violating triplets decays rapidly, reaching 96.5% zero-gradient triplets at steady state. The capacity to refine seen-class neighborhoods is therefore preserved, but the loss stops applying pressure once seen-class neighborhoods are locally separated, so the global reshaping that destroys unseen-class geometry under ICon and SRL never accumulates.

We evaluate EGA on CIFAR-100 (in-distribution) and three OOD benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), comparing against the frozen CLIP baseline, global contrastive adapters (ICo, SRL), and LoRA configurations with ranks $r = 64$ and $r = 128$ that are capacity-matched to EGA. The six methods occupy distinct operating points in the (ID, OOD) Label Precision plane. EGA reaches the highest in-distribution Label Precision among adapters that avoid OOD collapse, achieving 0.705 on CIFAR-100 versus 0.6176 for capacity-matched LoRA at $r = 128$, while bounding OOD degradation between -7 and -9 percentage points on CIFAR-10 and Food-101 (compared to between -32 and -33 percentage points for ICon and SRL) and improving by 9.9 percentage points on FGVC-Aircraft. The benefits transfer to stronger backbones, with EGA improving Label Precision@1 by 4.75 and 8.55 percentage points on DINOv2-large and SigLIP respectively. EGA does not dominate every method on every axis: on CIFAR-10 and Food-101, capacity-matched LoRA variants achieve higher OOD Label Precision than EGA, and deployments dominated by OOD queries on these distributions may prefer LoRA over EGA at the cost of weaker in-distribution refinement.

In summary, this work makes the following contributions:

- We frame adapter tuning over frozen vision-language encoders from a vector search deployment perspective, in which the index and the query stream together cover a label space larger than the labeled subset available for adapter training, and we characterize the failure modes of existing adapter paradigms under this lens.
- We propose Euclidean Geodesic Alignment (EGA), a residual adapter that combines a high-capacity zero-initialized architecture with a small-margin triplet objective. The triplet loss converges to a sparse-gradient regime (96.5% of training triplets produce zero gradient at steady state), which empirically preserves unseen-class geometry while permitting in-distribution refinement.

- Across CIFAR-100 and three OOD benchmarks, and across three backbones (CLIP ViT-B/32, DINOv2-large, SigLIP), EGA occupies a distinct operating point in the (ID, OOD) Label Precision plane, achieving the highest in-distribution Label Precision among adapters that avoid OOD collapse. We position this against capacity-matched LoRA variants, which preserve OOD precision better on two of three benchmarks at the cost of 8.7 percentage points in-distribution.

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [25] produce embedding spaces that encode both seen-class semantics and broader priors over unseen visual concepts. Catastrophic forgetting has been studied extensively in continual learning [20, 37], while recent work documents analogous degradation in contrastive representations, including anisotropic collapse [43, 6], modality gap fracturing [26, 5], and loss of compositional structure [24, 28, 36]. Recent strategies counter these issues with geometric constraints such as orthogonality regularization [27]. *EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen classes of the pretrained manifold unmodified.*

Global Contrastive Objectives and Structural Regularization Recent representation learning frequently relies on global contrastive objectives to optimize the latent space [31, 19]. A common paradigm jointly enforces alignment for positive pairs and uniformity across the feature distribution [34], a strategy applied in graph encoders [33] and multimodal recommendation [44]. While uniformity improves robustness to noise [35], it assumes uniform treatment of all latent regions. Auxiliary structural regularizers include unifying frameworks for multi-view alignment [1], geometric constraints for imbalanced regression [4], decoupled objectives [18], information-theoretic density control [42], and region-based distillation [15]. *EGA replaces dense global optimization with sparse local triplet updates, preserving out of distribution semantic structure.*

Parameter-Efficient Manifold Adaptation Parameter-efficient adaptation has become the standard approach to prevent catastrophic forgetting and reduce computational cost when transferring large foundation models [9, 10]. This paradigm now dominates vision-language research due to the prohibitive cost of full fine-tuning [21]. Lightweight methods such as Visual Prompt Tuning [14] and Tip Adapter [41] enable effective specialization without modifying the frozen backbone, and extensive benchmarks show they can rival full fine-tuning while preserving generalization [29]. Zero-initialization strategies further protect the pretrained manifold [40]. *EGA integrates a zero initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.*

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for scalable high-dimensional retrieval. The systems community has developed highly optimized indexing algorithms for billion-scale datasets, including FAISS [16], Hierarchical Navigable Small World graphs [22], DiskANN [30], and SPANN [2]. Recent work refines graph topologies [38, 3], optimizes SSD-based search [7, 8], and accelerates distance computations [13]. Nevertheless, popular ANN implementations still suffer from severe worst-case limitations [12]. *EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.*

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$. Our goal is to learn an adapter $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to unseen-class labels during training.

143 We evaluate retrieval quality along two complementary dimensions. *Label Precision@K* (LP@K)
 144 measures the semantic quality of retrieved results: whether the K nearest neighbors in the transformed
 145 space share the same class label as the query:

$$\text{LP@K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

146 where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k -th nearest neighbor of q in the transformed
 147 embedding space. *ANNS Recall@K* (AR@K) measures the geometric fidelity of the ANN index, i.e.,
 148 the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate
 149 index at a given search budget n_{probe} :

$$\text{AR@K} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

150 where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top- K neighbor sets, respectively.
 151 The two metrics capture distinct failure modes: an adapter may improve LP@K by tightening
 152 within-class clusters (semantic improvement) while simultaneously improving AR@K by making
 153 the geometry more amenable to IVF-based indexing (geometric improvement).

154 3.2 Manifold-Preserving Adapter Design

155 EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement
 156 layer. The architecture is constrained by three design principles, each of which serves a distinct role
 157 in preserving the pretrained manifold.

158 **(1) Residual connection with zero initialization** Each EGA block applies a residual update as:

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

159 where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely,
 160 the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This
 161 guarantees that training begins from the pretrained geometry rather than a random perturbation, and
 162 that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation
 163 (Table 4), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because
 164 the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

165 **(2) Feature gating** Inside each residual block, the transformed signal passes through a module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

166 where $W_1 \in \mathbb{R}^{(d/4) \times d}$, $W_2 \in \mathbb{R}^{d \times (d/4)}$, σ is the Sigmoid function, and $\odot$ denotes elementwise
 167 multiplication. This structure, analogous to Squeeze-and-Excitation networks [11], learns a per-
 168 dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones.
 169 The bottleneck ratio of 1/4 is chosen to prevent the gating network from overfitting on seen-class
 170 data, as validated in Table 5.

171 The output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 :

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

172 This constraint ensures that the transformed embeddings remain in the same metric space as the
 173 original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable
 174 on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage
 175 points (Table 4), confirming that hypersphere consistency is necessary for stable ANN indexing.

176 **(3) Full forward pass** Let Block_i denote the i -th EGA block consisting of a two-layer MLP
 177 with LayerNorm followed by feature gating, and let Refine denote the final linear projection with
 178 LayerNorm. The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is:

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z}))))). \quad (6)$$

179 The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension
 180 2048, which is negligible compared to the frozen CLIP backbone ($\sim 150M$ parameters).

3.3 Local Triplet Training and Gradient Sparsity

Given a mini-batch $\mathcal{B}$ sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a is an anchor image, p is a positive sampled uniformly from the same class as a , and n is a negative sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the transformed embedding.

A triplet (a, p, n) contributes a non-zero gradient to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many triplets satisfy the margin constraint and become inactive, contributing zero gradient. This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided that the adapter’s updates remain effectively local (i.e., the cumulative perturbation induced by seen-class training gradients remains small in unseen-class regions of the embedding space), the manifold structure for unseen classes is largely preserved. The gradient sparsity mechanism enforces this locality: once seen-class neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of actively-violated local relationships continue to be updated.

We formalize these intuitions in the following observation, which characterizes the steady-state behavior of triplet-trained adapters and its implications for manifold preservation.

Gradient sparsity at convergence. The triplet loss in Eq. (7) consists of hinge terms whose subgradient with respect to θ is exactly zero whenever the margin constraint is satisfied:

$$\nabla_\theta \ell_{apn} = \mathbf{1}[d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0] \cdot \nabla_\theta \delta_{apn}, \quad (9)$$

where $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. Define the *active triplet ratio* at training step t as

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} [d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0]. \quad (10)$$

Under the standard assumption that training converges to a stationary point of $\mathcal{L}$, $\rho(t)$ converges to a steady-state value $\rho^* \in [0, 1]$ determined by the margin m and the within-class distance distribution of the converged embeddings. We measure $\rho^* \approx 3.5\%$ at convergence on FGVC-Aircraft (Section 4.3), meaning that 96.5% of training triplets contribute zero gradient at steady state.

From sparsity to OOD locality: an empirical claim. Strictly speaking, sparse seen-class gradients do not formally imply that f_θ leaves unseen-class regions unchanged: the adapter is a parametric MLP, not a locally supported transformation, so any update to θ propagates globally. We therefore make the weaker, empirical claim that the residual zero-initialized architecture combined with gradient sparsity keeps the cumulative perturbation in unseen-class regions small enough to preserve their pretrained semantic structure for retrieval. The OOD measurements in Section 4.2 are the evidence: EGA preserves or improves Label Precision on three unseen-class benchmarks, while ICon and SRL lack the gradient sparsity property and fall below the frozen baseline. A formal Lipschitz-based bound on the unseen-class perturbation is left for future work; we discuss the relevant technical conditions in Appendix A.

4 Evaluation

Experiments were conducted on the Chameleon Cloud platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in distribution scenarios, CIFAR-10 for cross dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen class OOD evaluations. All of these datasets can be directly downloaded from the Pytorch library. The backbone models include CLIP ViT-B/32 [25], DINOv2-large [23], and SigLIP [39]. We compare our method against four adapter models: ICon [1], SRL [4], LoRA + InfoNCE [10, 32], and LoRA + Triplet [10].

Extended experimental details can be found in Appendix B.

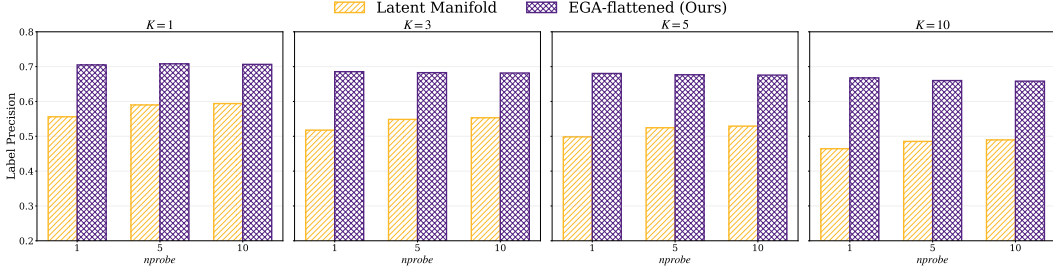


Figure 1: Label Precision@ K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and $nprobe \in \{1, 5, 10\}$.

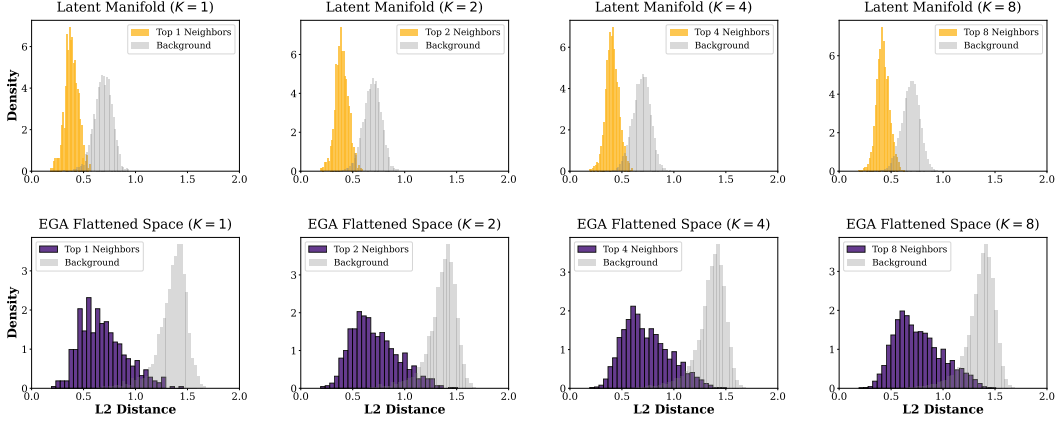


Figure 2: Distance distributions for Top- K neighbors vs. random background points on CIFAR-100.

Table 1: In-distribution retrieval performance on CIFAR-100 at $K = 1$, $nprobe = 1$.

Metric	CLIP	ICoN	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
LP@1	0.549	0.999	0.992	0.668	0.612	0.705
AR@1	0.667	0.992	0.991	0.879	0.908	0.825

225 4.1 In-Distribution Validation

226 Figure 1 reports Label Precision@ K across four neighborhood sizes and three search budgets. EGA
 227 improves Label Precision@1 from 0.549 (raw CLIP) to 0.705 at $nprobe = 1$, and the gap persists
 228 across all K and $nprobe$ values. The improvement is not an artifact of larger search budgets: even at
 229 $nprobe = 10$, EGA retains a clear advantage: the adapter tightens within-class neighborhoods rather
 230 than benefiting from broader index coverage.

231 The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance
 232 distributions for true Top- K neighbors against random background points, before and after EGA. In
 233 the raw CLIP space, the two distributions overlap substantially: the distance to a true neighbor is
 234 often indistinguishable from the distance to an unrelated point, which is exactly the condition under
 235 which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across
 236 all K , with neighbors concentrated at small distances and background points pushed toward a distinct
 237 mode. This separation is the mechanism by which EGA improves the recall-efficiency frontier shown
 238 in Figure 3: the raw CLIP space achieves only 0.667 Recall@1 at $nprobe = 1$, whereas EGA reaches
 239 0.825 and saturates above 0.99 by $nprobe = 5$.

240 While Figures 1–3 focus on the EGA-vs-frozen contrast, we also report all six methods’ in-distribution
 241 performance in Table 1, since the contrast with their out-of-distribution behavior (Section 4.2) is
 242 central to our analysis. ICoN and SRL achieve near-perfect Label Precision (0.999 and 0.992
 243 respectively) by maximally tightening seen-class clusters with their global contrastive objectives.
 244 EGA reaches LP@1 of 0.705, an improvement of +16 pp over the frozen baseline; the LoRA variants

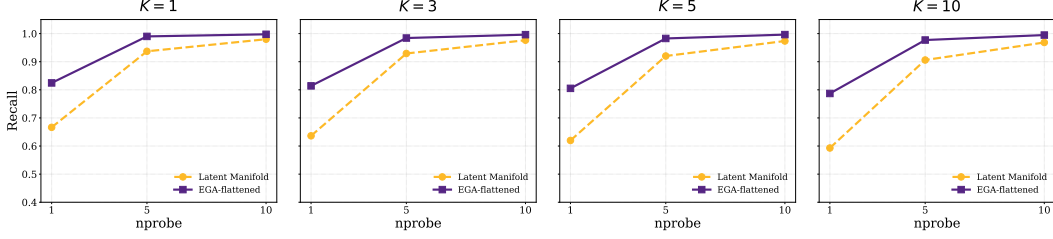


Figure 3: ANNS Recall@ K versus nprobe on CIFAR-100.

Table 2: OOD performance at $K = 1$, nprobe = 1, mean $\pm$ std over 3 random seeds (42, 123, 456).

Dataset	Metric	CLIP	Icon	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	<u>0.797</u> $\pm$.003	0.819 $\pm$.001	0.869 $\pm$.012	0.861 $\pm$.011	0.833 $\pm$.005
	LP@1	0.880	<u>0.560</u> $\pm$.010	0.536 $\pm$.008	0.885 $\pm$.006	0.875 $\pm$.006	0.810 $\pm$.002
Aircraft	AR@1	0.773	0.853 $\pm$.016	0.879 $\pm$.023	0.891 $\pm$.015	0.893 $\pm$.008	0.905 $\pm$.015
	LP@1	0.512	<u>0.470</u> $\pm$.034	0.375 $\pm$.032	0.538 $\pm$.020	0.569 $\pm$.011	0.611 $\pm$.020
Food-101	AR@1	0.842	<u>0.821</u> $\pm$.001	0.824 $\pm$.015	0.906 $\pm$.003	0.893 $\pm$.008	0.879 $\pm$.011
	LP@1	0.881	<u>0.562</u> $\pm$.011	0.552 $\pm$.010	<u>0.883</u> $\pm$.004	<u>0.833</u> $\pm$.007	0.791 $\pm$.002

land between EGA and the baseline. Section 4.2 will show the same property allowing Icon and SRL to dominate this in-distribution metric is what causes their catastrophic out-of-distribution collapse.

4.2 Out-of-Distribution Generalization

Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories about adapter quality, and EGA is the only method that performs well on both. Table 2 reports headline numbers at $K = 1$, nprobe = 1. Figure 4 visualizes the joint (ID, OOD) tradeoff across all six methods, and Appendix C verifies that the pattern is robust across search budgets (nprobe $\in \{1, 5, 10\}$). We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions.

On ANNS Recall, the adapters either match or exceed the frozen CLIP baseline on Aircraft and Food-101, indicating that adapter training succeeds at geometric reorganization on these two benchmarks. On Label Precision, however, the picture inverts. Icon and SRL fall *below* the frozen CLIP baseline on every dataset, with degradation reaching -32 to -33 percentage points on Food-101. The retrieved neighbors are geometrically closer in the IVF index, but they belong to the wrong semantic class. EGA improves Label Precision over the frozen baseline on Aircraft (+9.9 pp), with bounded degradation on CIFAR-10 (-7.0 pp) and Food-101 (-9.0 pp). The corresponding collapse for Icon and SRL is much larger: -32 to -33 pp on Food-101 and -32 to -34 pp on CIFAR-10.

EGA’s relative advantage over global methods ranges from 14 pp on Aircraft to 23 pp on Food-101 and 25 pp on CIFAR-10, and tracks the difficulty of the OOD shift: when seen and unseen classes are visually similar (as in Food-101’s fine-grained dishes), the dense gradient updates of Icon and SRL pull unseen samples most aggressively into the wrong seen-class clusters. Robustness across search budgets (nprobe $\in \{1, 5, 10\}$) is reported in Appendix C.

Cross-referencing Table 1 and Table 2 reveals a structural tradeoff among adapter training paradigms. Icon and SRL achieve near-perfect Label Precision on CIFAR-100 (0.999 and 0.992) but collapse to 0.562 and 0.552 on Food-101. The frozen baseline and LoRA variants sit at the opposite extreme: modest in-distribution precision (0.55 to 0.67) but stable performance under distribution shift. EGA occupies a balanced position: 0.705 in-distribution and 0.791 on Food-101, with 0.611 on Aircraft and 0.810 on CIFAR-10.

The six methods we evaluate trace out a Pareto frontier in the (ID Label Precision, OOD Label Precision) plane, in Figure 4. The dashed line interpolates between non-dominated points (those for which no other method achieves both higher ID and higher OOD Label Precision). Icon and SRL achieve the highest in-distribution precision but collapse on OOD; the LoRA variants achieve

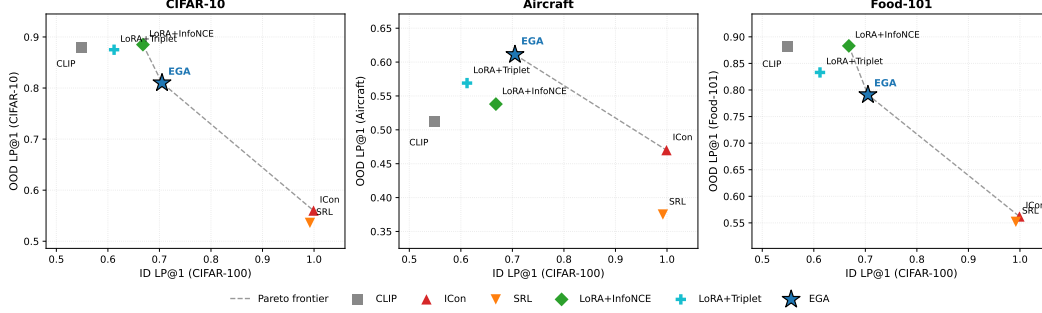


Figure 4: ID/OOD Pareto frontier of adapter training paradigms.

Table 3: Performance of EGA versus capacity-matched LoRA ($K = 1$, $nprobe=1$).

Dataset	Metric	EGA	InfoNCE r=64	InfoNCE r=128	Triplet r=64	Triplet r=128
CIFAR-100 (ID)	LP@1	0.705	0.6256	0.6176	0.5568	0.5472
	AR@1	0.825	0.879	0.879	0.908	0.908
Aircraft (OOD)	LP@1	0.611	0.5417	0.5298	0.5833	0.5417
	AR@1	0.905	0.6190	0.6131	0.6667	0.6726
Food-101 (OOD)	LP@1	0.791	0.8667	0.8599	0.8599	0.8568
	AR@1	0.879	0.6504	0.6702	0.6756	0.6740
CIFAR-10 (OOD)	LP@1	0.810	0.8788	0.8728	0.8768	0.8756
	AR@1	0.833	0.6008	0.5940	0.6216	0.6096

competitive OOD precision on Food-101 and CIFAR-10 but underperform EGA in-distribution by 4 to 9 percentage points. EGA occupies a distinct operating point on this frontier, combining the highest in-distribution precision among adapters that avoid OOD collapse with bounded OOD degradation.

Table 3 shows that EGA outperforms capacity-matched LoRA ($r=64$ and $r=128$) on in-distribution Label Precision while maintaining competitive out-of-distribution performance. On CIFAR-100, EGA achieves 0.705 LP@1, exceeding the best capacity-matched LoRA by 7.9 percentage points. Across OOD benchmarks, EGA either matches or exceeds LoRA on Aircraft and remains within 7–9 percentage points on Food-101 and CIFAR-10. LoRA’s in-distribution performance slightly decreases with increasing rank, which we attribute to the low-rank constraint acting as an implicit regularizer.

4.3 Two Mechanisms for OOD Stability

The OOD pattern in Section 4.2 raises a mechanistic question: why do some adapters preserve unseen-class structure while others reshape it under global contrastive objectives? As we will show, experiments suggest two distinct mechanisms for OOD stability, which place adapters at different operating points on the Pareto frontier. Formally, a triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

As the adapter converges on seen classes, the fraction of *active* triplets decays. Figure 5 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by convergence. At steady state, 96.5% of triplets contribute zero gradient, meaning the adapter updates a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space intact.

Global contrastive objectives such as ICon [1] and SRL [4] exhibit the opposite behavior. ICon minimizes a Kullback–Leibler (KL) divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure shapes the entire embedding distribution based on seen-class similarities, even for regions of the space occupied by unseen-class

Table 4: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 5: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

samples. The result is not loss of geometric structure but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward other unseen samples that share their true category.

Figure 5 (right) and Table 2 together quantify the consequence: across all three OOD benchmarks, ICon and SRL fall below the frozen CLIP baseline on Label Precision (e.g., on Aircraft, ICon collapses to 0.470 from a baseline of 0.512), confirming that combining dense gradient fields with high adapter capacity harms zero-shot semantic retrieval. EGA, by contrast, improves Label Precision over the frozen baseline on Aircraft and bounds the degradation on the other two OOD benchmarks, while only 3.5% of training triplets contribute non-zero gradients at convergence.

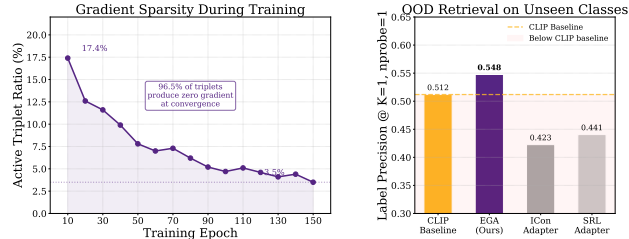


Figure 5: Gradient sparsity as OOD protection.

4.4 Ablation and Sensitivity Analysis

Table 4 isolates the contribution of each architectural component. The residual connection is the core element. Removing it causes accuracy to collapse from 0.7015 to 0.0065. This indicates that the adapter cannot reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. The L2 normalization ensures the transformed embeddings remain on the unit hypersphere. Removing it decreases accuracy to 0.6590. Zero-initialization allows the adapter to begin training as an identity mapping; removing it lowers accuracy to 0.6840.

Table 5 evaluates model stability across hyperparameter variations. EGA maintains consistent quality across margins from 0.1 to 0.3. Increasing the margin to 0.5 degrades Recall@1 to 0.7400. A larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, which reduces gradient sparsity and alters the pre-trained manifold. Furthermore, the model shows stable performance across hidden dimensions from 512 to 4096. This stability confirms that EGA achieves metric correction through its structural design rather than specific parameters.

4.5 Generalization to More Backbones

To verify that the benefits of EGA are not specific to the CLIP ViT-B/32 backbone used in our main experiments, we further evaluate EGA on two stronger and more recent vision encoders: DINOv2-large [23] and SigLIP [39].

As shown in Table 6, EGA delivers consistent improvements across all three backbones. On the relatively weaker CLIP ViT-B/32, EGA achieves substantial gains of +13.95 percentage points in Label Precision@1 and +13.20 percentage points in Recall@1. Even on the much stronger DINOv2-large and SigLIP backbones, EGA continues to provide meaningful and stable improvements (+4.75pp and +8.55pp in Label Precision@1, respectively). These results demonstrate that

Table 6: Retrieval performance on more backbones.

Backbone	Label Precision@1		ANNS Recall@1	
	Raw	+EGA	Raw	+EGA
CLIP ViT-B/32	0.5560	0.6955	0.6665	0.7985
DINOv2-large	0.8530	0.9005	0.8785	0.9310
SigLIP SO400M	0.7740	0.8595	0.8015	0.9155

348 EGA’s manifold-preserving design generalizes robustly to modern, high-capacity vision models and
349 is not merely an artifact of the original CLIP representation.

350 5 Conclusion

351 We approached adapter training from a vector search deployment perspective, where the index and
352 query stream together cover a label space larger than the labeled subset available at adapter training
353 time. Under this lens, adapter training does not produce uniformly improved retrieval embeddings.
354 Different paradigms occupy different operating points in the (ID, OOD) Label Precision plane, and
355 the choice among them reflects which slice of the deployed query distribution one is willing to serve
356 poorly.

357 Within this space, EGA offers a specific tradeoff: 0.705 in-distribution Label Precision on CIFAR-
358 100, with OOD degradation bounded at -7 to -9 percentage points on CIFAR-10 and Food-101
359 and an improvement of $+9.9$ percentage points on FGVC-Aircraft. ICon and SRL achieve higher
360 in-distribution numbers (0.999, 0.992) but fall 32 to 33 percentage points below the frozen baseline
361 on Food-101; capacity-matched LoRA ($r = 128$) preserves OOD precision better on two of three
362 benchmarks but trails EGA by 8.7 percentage points in-distribution. The benefits transfer to stronger
363 backbones: on DINOv2-large and SigLIP, EGA improves Label Precision@1 by 4.75 and 8.55
364 percentage points respectively. The mechanism we associate with EGA’s behavior is gradient sparsity
365 at convergence (96.5% of training triplets produce zero gradient), which we suggest as one useful
366 diagnostic for adapter designs intended to operate under unseen-class queries.

367 **Limitations.** Three limitations are worth stating explicitly. First, our analysis of gradient sparsity
368 (Observation ??) is empirical rather than formal: parameter updates from seen-class triplets affect f_θ
369 globally, and we do not provide a Lipschitz-based bound on the resulting perturbation in unseen-class
370 regions. The locality property is supported by OOD measurements, not by analysis. Second, EGA
371 does not dominate every competitor on every axis. On CIFAR-10 and Food-101, capacity-matched
372 LoRA variants achieve higher OOD Label Precision; deployments dominated by OOD queries on
373 these distributions may prefer LoRA over EGA. Third, our experiments hold the loss family and
374 architecture varying together in some comparisons, which prevents a clean disentanglement of how
375 much of EGA’s behavior is attributable to the triplet objective versus the residual gating architecture.
376 Controlled sweeps over capacity and loss under matched conditions are the natural next step.

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A Technical Conditions for the Sparsity-Locality Claim

This appendix elaborates on the technical conditions underlying the empirical claim made in Section 3.3.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_{\theta} \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_{θ} and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to ρ^* as defined in Eq. (??), since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Empirical locality. Strictly speaking, parameter updates $\nabla_{\theta} \delta_{apn}$ computed on seen-class triplets affect f_{θ} globally (including its behavior on unseen-class inputs) since f_{θ} is a parametric MLP and is not locally supported on the input manifold. We therefore do not claim formal zero-gradient over unseen regions. Instead, we observe an empirical locality property: when (i) the residual structure with zero initialization keeps the cumulative perturbation magnitude bounded by the sum of training gradients, and (ii) gradient sparsity at convergence ($\rho^* \approx 3.5\%$ on FGVC-Aircraft) attenuates this sum, the resulting perturbation in unseen-class regions is small enough to preserve the pretrained semantic structure for retrieval purposes. This observation is empirically supported by the OOD Label Precision results in Section 4.2 and Figure 6, where EGA preserves or improves Label Precision on unseen categories while ICon and SRL, which lack the gradient sparsity property, collapse below the frozen baseline. A formal Lipschitz-based bound on this perturbation is left for future work.

(4) Dependence on m . The residual distance gap δ_{apn}^* at convergence is bounded by the within-class diameter $\Delta_c = \max_{x, x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c . For $m \leq \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^* . This explains the empirically observed increase in ρ^* with larger m reported in Table 5.

B Experimental Setup Details

Backbones models. All experiments in the main paper use frozen CLIP ViT-B/32 [25] image embeddings as the base representation. To verify that EGA’s benefits are not specific to this backbone, we further evaluate EGA on two stronger vision encoders: DINOv2-large [23] and SigLIP [39]. DINOv2-large is a self-supervised vision transformer trained on 142M images, while SigLIP is a supervised vision transformer trained on 400M image-text pairs. Both models produce higher-quality embeddings than CLIP ViT-B/32, as reflected in their stronger raw retrieval performance, and EGA continues to provide meaningful improvements on top of these stronger backbones, as shown in Table 6 in Section 4.4.

Adapter models. We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **ICOn** [1]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution. We instantiate ICon on the same EGAMLP architecture used by our method ($\sim 4.2\text{M}$ parameters).
- **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.
- **LoRA + InfoNCE** [10, 32]: a capacity-matched low-rank adapter (rank $r = 64$ and $r = 128$, $\sim 2.6\text{M}$ – 4.2M parameters) with zero-initialized residual structure, trained with supervised InfoNCE, which is a standard global contrastive loss.

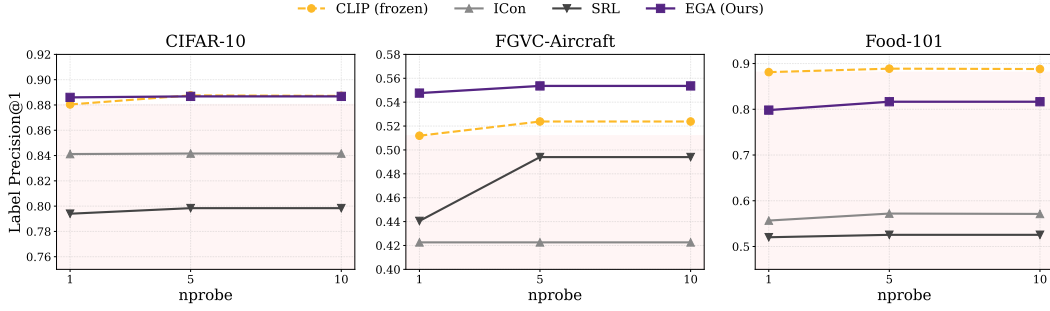


Figure 6: Label Precision@1 across three OOD benchmarks and three search budgets ($nprobe \in \{1, 5, 10\}$).

- **LoRA + Triplet** [10]: identical capacity-matched low-rank architecture as above (rank $r = 64$ and $r = 128$), but trained with the same small-margin ($m = 0.2$) triplet loss as EGA.

The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing LoRA + InfoNCE against ICoN/SRL holds the loss family roughly fixed while changing capacity, and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

Datasets. We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained and evaluated on the same class label space, with a 75/25 split between database and query sets. **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

Indexing and metrics. For all retrieval evaluations we use FAISS [16] IndexIVFFlat with $nlist=10$ centroids and report results at $nprobe \in \{1, 5, 10\}$. We measure both Label Precision ($LP@K$, Eq. 1) and ANNS Recall ($AR@K$, Eq. 2) at $K \in \{1, 3, 5, 10\}$.

Training. All adapters are trained for 150 epochs with AdamW (weight decay 10^{-4}), cosine-annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICoN, SRL) use learning rate 10^{-4} ; LoRA variants use 10^{-3} to compensate for their smaller parameter count. Triplet-based methods sample one positive and one negative per anchor uniformly within each mini-batch; global contrastive methods (ICoN, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix. All experiments use a fixed random seed (42) for reproducibility.

Compute platform. All experiments were conducted on the Chameleon Cloud [17] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.

C Additional Experimental Results

Robustness across search budgets. Figure 6 traces Label Precision@1 across three OOD benchmarks and three search budgets ($nprobe \in \{1, 5, 10\}$). The collapse pattern of ICoN and SRL persists across all 18 measurement points (3 datasets $\times$ 3 budgets $\times$ 2 methods), ruling out the possibility that the degradation is an artifact of any particular benchmark or indexing configuration. EGA’s relative ordering against other methods is also stable across budgets.